



Pharm*Achieve*

PHARMACOECONOMICS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

CBA	Cost-Benefit Analysis
CE	Cost-Effectiveness
CEA	Cost-Effectiveness Analysis
CIA	Cost-Illness Analysis
CMA	Cost-Minimization Analysis
COPD	Chronic Obstructive Pulmonary Disease
CUA	Cost-Utility Analysis
FYI	For Your Information
ICEA	Incremental Cost-Effectiveness Analysis
ICER	Incremental Cost Effectiveness Ratio
LDL	Low Density Protein
MI	Myocardial infarction
PEBC	Pharmacy Examining Board of Canada
QALYs	Quality Adjusted Life Years
SA	Sensitivity Analysis
WTP	Willingness-To-Pay

NOTE TO READER

Anything marked “**FYI**” in this slide deck has, in PharmAchieve’s view, a LOW chance of being on the exam, therefore we do not recommend you expend significant time on concepts marked FYI.

Note: “low chance” does NOT mean “zero chance”; it could still be on your exam.

INTRODUCTION: PHARMACOECONOMICS IS A BRANCH OF OUTCOMES RESEARCH



Outcomes Research
(identify, measure & evaluate end results of healthcare services)

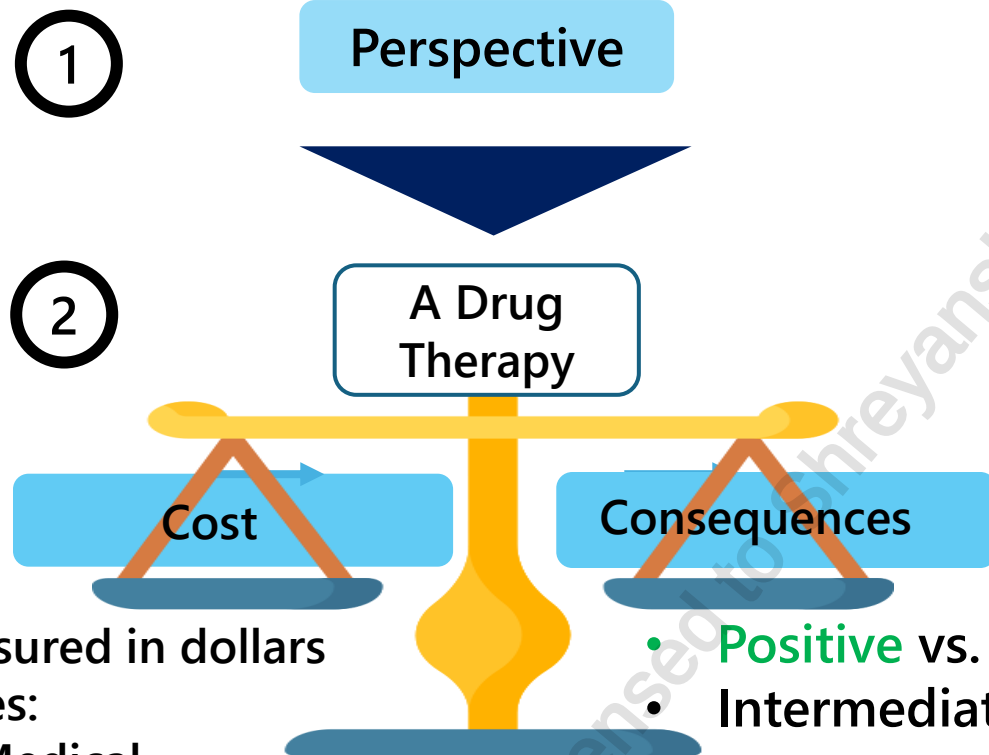
The diagram consists of two concentric circles. The outer circle is a medium blue color and contains the text 'Outcomes Research (identify, measure & evaluate end results of healthcare services)'. The inner circle is a darker blue color and contains the text 'Pharmacoeconomics (identifies, measures and compares cost and consequences of drug therapies)'. This visualizes pharmacoeconomics as a specialized branch within the broader field of outcomes research.

Pharmacoeconomics
(identifies, measures and compares cost and consequences of drug therapies)

Pharmacoeconomics $\neq$ outcome research

PHARMACOECONOMICS

Once a perspective is chosen, the cost and consequences of choosing a product can be measured using pharmacoeconomic methods



Cost - Measured in dollars

4 Categories:

1. Direct Medical
2. Direct Non-Medical
3. Indirect
4. Intangible

- Positive vs. negative
- Intermediate vs. final

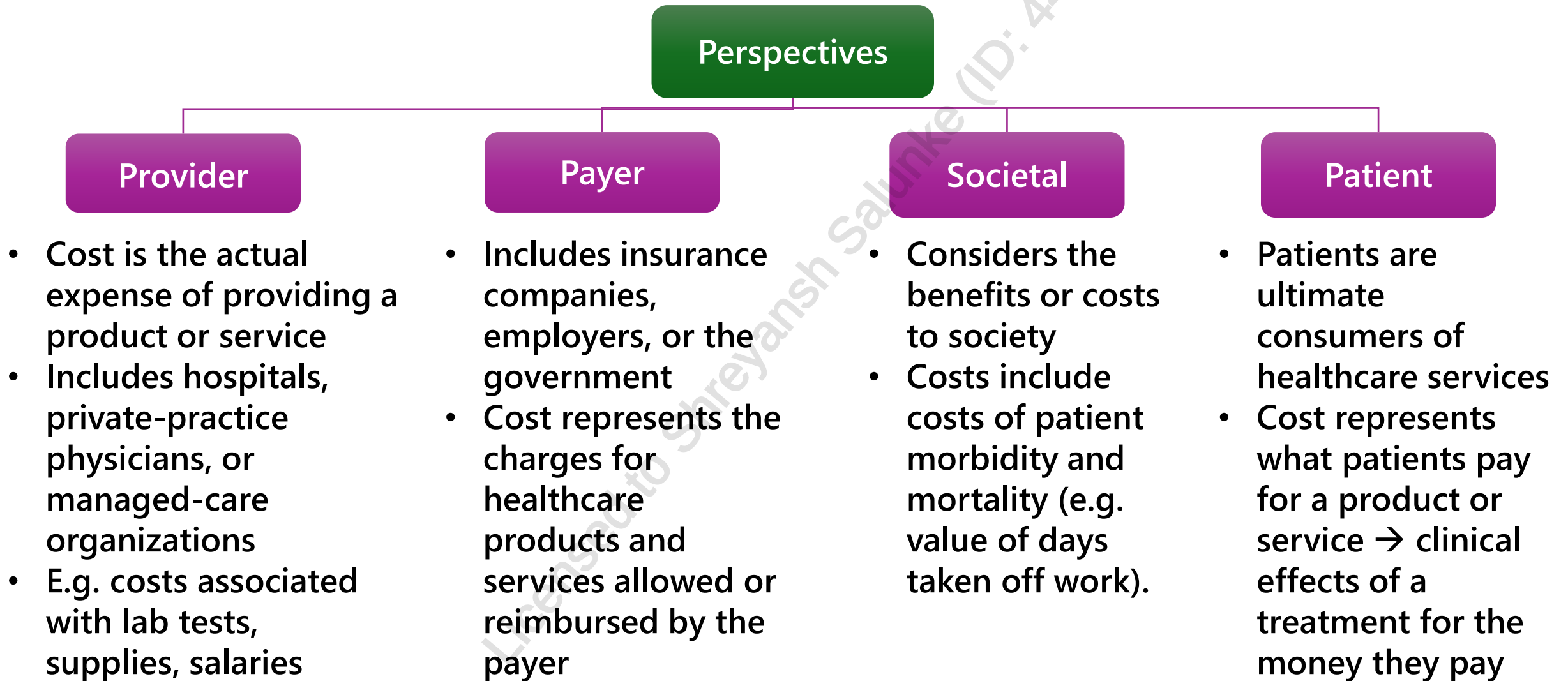
Categories of consequences

- 1 Economic (expense, savings, budget impact, cost, avoidance, cost of care)
- 2 Clinical (safety, efficacy, cure, mortality, morbidity, hospital stay)
- 3 Humanistic (health quality of life, patient satisfaction)

4 ways to measure consequences (outcomes):

- Cost-Minimization Analysis (CMA)
- Cost-Benefit Analysis (CBA)
- Cost-Effectiveness Analysis (CEA)
- Cost-Utility Analysis (CUA)

PRINCIPLES OF PHARMACOECONOMICS



PRINCIPLES OF PHARMACOECONOMICS: COST

Cost categories	Definitions	Examples
Direct medical costs	Cost of medical products/services to prevent, detect, or treat a disease	Medications, supplies, lab tests, health care professional's time, hospitalization, physician visits
Direct non-medical costs	Cost associated with non-medical services due to illness or disease	Transportation to healthcare facilities, food, family care, home aides
Indirect non-medical costs	Costs of reduced productivity from illness or death	Cost of Illness Study (COI): outlining total cost of Morbidity (value of work lost owing to illness), or mortality (value of work lost due to premature death).
Intangible Costs	Cost of pain, suffering, grief, anxiety or fatigue resulting from an illness	How much patient would be willing to pay to alleviate their pain and anxiety* Estimated through WTP

PRINCIPLES OF PHARMACOECONOMICS

Willingness-To-Pay approach (WTP):

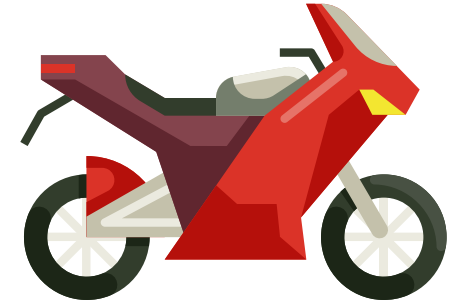
- Estimates indirect and intangible costs
- Subjective – therefore results are unreliable
- Asks patients how much money they would be willing to spend to reduce likelihood of illness

Cost categories	Definitions	Examples
Intangible costs	Costs that are difficult to measure quantitatively	Pain, suffering, inconvenience, grief
Opportunity costs	Benefit foregone when using one therapy instead of the next best alternative therapy	See next slide
Incremental costs	Additional costs and benefits arising as a consequence of using a treatment alternative over another	See next slide

PRINCIPLES OF PHARMACOECONOMICS

Understanding the opportunity cost:

- Suppose you can either buy a motorcycle or go on a vacation but not both
- The opportunity cost of buying a motorcycle is your inability to enjoy the benefits of going on a vacation
- In health care system, this concept is important because we have to analyze the health benefits that would arise from using resources that are currently dedicated to other services



Understanding the incremental costs:

- Suppose a new medication is now available for the management of COPD
- The incremental cost analysis would examine the **additional costs and benefits** of using the new drug in comparison to the existing ones

PRINCIPLES OF PHARMACOECONOMICS: CONSEQUENCES FYI

Three categories:

Economic	Clinical	Humanistic
Expense, savings, cost, avoidance: cost associated with care such as hospital readmission	Cure, comfort and survival (safety, efficacy), mortality/morbidity, hospital stay	Patient physical, emotional, social functional status (health quality of life) or quality of life, patient satisfaction

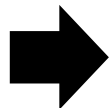
Outcomes of a disease are an important component of pharmacoeconomic analyses



These can be further divided into positive outcomes and negative outcomes

- **Positive**: desired effect of a drug, life-years gained, improved quality of life
- **Negative**: undesired or adverse effect of a drug, toxicity

Intermediate vs. final consequences:



- E.g. lowering of LDL with a lipid-lowering agent is an intermediate consequence, whereas reduction in MI rate is a final consequence

QUALITY ADJUSTED LIFE YEAR

How to interpret Quality Adjusted Life Years (QALYs):

1 QALY = 1 year of perfect health



Years of less than perfect health are accrued at less than 1 QALY per year
e.g. if you were 50% healthy, you could accrue QALYs at 0.5/year



Upon death QALY are accrued at 0 per year

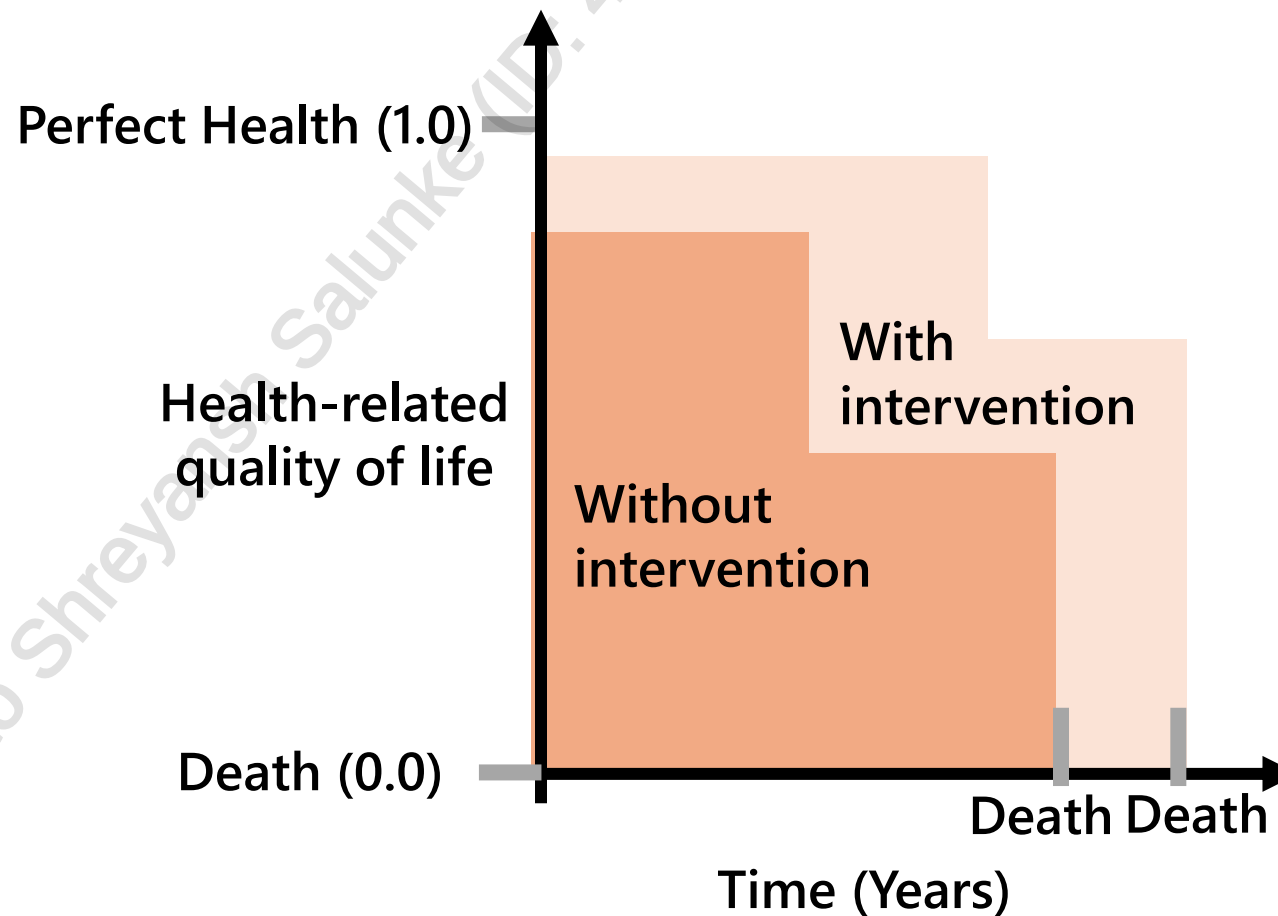


It is even possible to have negative QALY for years when the patient's
quality of life is "worse than death"

QUALITY ADJUSTED LIFE YEAR

Sometimes patients need to choose between different options such as treatments or not treatment at all. How can they decide?

One intermediate step in making a decision is calculating the Quality Adjusted Life Years (QALYs) for each option



HOW DO YOU ESTIMATE QALY?

Health Utilities:

- Utility is a measure of desirability or goodness
- Several ways to measure utility for a health outcome:
 - Patients are given different health outcomes
 - Patients are then asked to rate these health outcomes on a rating scale usually between 0 and 1 with 0 often representing death and 1 representing best possible health

Health utilities are often combined with survival estimates and aggregated across individuals to estimate QALYs

- These QALYs are not for a specific patient, but are often population averages based on survey of some portion of the public (usually existing patients)
- These QALY estimates are used in CUA Analysis; for instance, to decide which procedures should be funded in a public health system, where not everything can be funded and choices must be made

QUALITY ADJUSTED LIFE YEAR

1. Rating scale / visual analogue approach

- Patients are given a line with anchor ranging from death (representing 0) to best possible health (representing 1.0)
- They are asked to place different health outcomes (e.g. bedridden, able to walk with some assistance, etc.) on the line
- *Example:* patient places their current condition at 0.7. This means in their current condition they accrue 0.7 QALY.

QUALITY ADJUSTED LIFE YEAR

There are three basic approaches to measure QALYs for a patient:

2. Time-trade off approach

- Most common method used to arrive at quality-adjust life years
- Asks patients the years of life they are willing to sacrifice to live remaining years in perfect health
- *Example:* a patient is asked the following: "Imagine that you are told that you have 10 years left to live and that you can choose to (1) live these 10 years in your current health state; or (2) give up some life years to live for a shorter period but in perfect health. Indicate with a cross on the line the number of years in full health that you think is of equal value to 10 years in your current health state."
- If patient puts a cross at 6 years, then they are saying that 10 years at their current state of health is worth 6 years in perfect health (so 1 year in current state = 0.6 years in perfect health, so QALY in current state accrue at 0.6/year)

QUALITY ADJUSTED LIFE YEAR

3. Standard gamble approach



- Suppose you wished to assess the QALY that a patient would assign to one year with a certain disease condition (e.g. frequent migraines)
 - Under this approach, you provide the patient with a theoretical choice to continue with the current disease condition (e.g. frequent migraines) or to “gamble”. If they win, the migraine goes away; if they lose, they die
 - Ask what probability the patient would need for the “good” outcome before they equally prefer the gamble to their current state
 - The gamble consists of a X% chance of getting perfect health or a 100-X% chance of death
 - You begin with an X of 1% and increase it in 1% intervals to see the % where the decision to accept vs reject gamble changes from rejecting to being indifferent:
 - Example: When chance of winning is ..
 - **1 to 79%**: patient prefers to keep their migraines
 - **80%**: patient equally prefers to gamble vs. continuing to live with migraines (this is the change in decision we are looking for)
 - **> 81%**: patient prefers the gamble
 - This means that in their current situation, each year of life equals **0.8** QALYs in perfect health

PHARMACOECONOMIC METHODS: COST-MINIMIZATION ANALYSIS (CMA)

- Choosing the least costly alternative when comparing several treatment options with the same efficacy and safety – only cost is considered
- Shows cost savings of one drug over another
- Cannot be used to evaluate therapies that have different outcomes
- E.g. CMA is used when hospitals decide to introduce generic drugs instead of their brand names
- Note: there is one such question in the sample questions provided by the PEBC
- Note: if you have two “best” options – both with the lowest cost, pick the option with the least number of doses required (example: pick BID instead of QID).

PHARMACOECONOMIC METHODS

Cost-Minimization Analysis - Practice question:

If Drugs A, B, and C have the same efficacy and safety, which drug is the best alternative?

	Administration	Cost per pill	Monitoring cost
Drug A	QID for 10 days	\$0.25	\$70
Drug B	BID for 20 days	\$1.25	\$50
Drug C	Daily for 20 days	\$1.50	\$55

Answer:

Drug A: $(4 \times 10 \times 0.25) + 70 = \80

Drug B: $(2 \times 20 \times 1.25) + 50 = \100

Drug C: $(1 \times 20 \times 1.50) + 55 = \85

Based on CMA, drug A is the least costly alternative!

PHARMACOECONOMIC METHODS: COST-BENEFIT ANALYSIS (CBA)

- Identifies and measures the total costs of providing a treatment or a program and compares the cost to the benefit that results from using such programs/treatments
- Benefits are given a monetary value so cost and benefits can be compared, allowing totally different interventions to be easily compared

$$\text{Benefit to Cost ratio} = \frac{\text{benefit (\$)}}{\text{cost (\$)}}$$

- The greatest **B:C ratio** is the best alternative
- If **B:C > 1** → Program or treatment is of value, benefit exceeds cost
- If **B:C = 1** → Benefit of program/treatment is equivalent to the cost of providing it
- If **B:C < 1** → Cost exceeds benefit, not economically beneficial
- Since it is hard to measure indirect and intangible benefits and costs (such as patient discomfort, risk of adverse effects), CBAs are used to quantify direct costs

PHARMACOECONOMIC METHODS: COST-BENEFIT ANALYSIS (CBA)

Practice question: based on CBA, which drugs is the best to use?

	Administration	Cost per day	Benefit
Drug A	QID for 10 days	\$100	\$1500
Drug B	BID for 20 days	\$5	\$100
Drug C	Daily for 30 days	\$400	\$6000

Answer:

Drug A: $\$1500 / (\$100 \times 10) = 1.5$

Drug B: $\$100 / (\$5 \times 20) = 1.0$

Drug C: $\$6000 / (\$400 \times 30) = 0.5$

Drug A's B:C ratio is the highest and is > 1.0 so it is the best alternative

Drug B's B:C is 1.0 so its benefits only match the costs

Drug C's B:C is 0.5 so its costs exceed the benefits (and should not be used)

PHARMACOECONOMIC METHODS: COST-EFFECTIVENESS ANALYSIS (CEA)

- Compares alternatives with different safety and efficacy profiles (for treating the same condition) by weighting cost against efficacy
- Allows costs and outcomes to be reduced to a single value for comparison
- Cannot be used to compare different areas of medicines with different outcomes

$$\text{Cost-Effectiveness} = \frac{\text{health care costs (\$)}}{\text{clinical outcome (in measurable units, not \$)}}$$

Examples:

- \$/life saved
- \$/cases cured
- \$/mmHg drop in BP, etc.

Note: you want to pay the least possible amount for the effect, so ... lower CE Ratios are better

PHARMACOECONOMIC METHODS: COST-EFFECTIVENESS ANALYSIS

CEA Practice question: Which of the following blood pressure medications with different safety and efficacy profile is the most cost-effective to use? (numbers shown over 1 month period)

	Administration	Cost per month	Drop in mmHg
Drug A	QID	\$100	10
Drug B	BID	\$200	5
Drug C	Daily	\$400	8

Answer:

Drug A: $\$100 / 10\text{mmHg} = \$10 / \text{Drop in mmHg}$
Drug B: $\$200 / 5\text{mmHg} = \$40 / \text{Drop in mmHg}$
Drug C: $\$400 / 8\text{mmHg} = \$50 / \text{Drop in mmHg}$

Based on CEA, Drug A is associated with the lowest cost per mmHg drop. Therefore, it is the most cost-effective agent to use.

PHARMACOECONOMIC METHODS: INCREMENTAL CEA (ICEA)

FYI

Also common is the **Incremental ECA (ICEA)**. It is used to determine the additional cost of one additional unit of benefit

$$\text{Incremental CEA (ICEA)} = \frac{\text{Cost A} - \text{Cost B}}{\text{Effect A} - \text{Effect B}}$$

Example: Hypertension Drug A is superior to Drug B in that it achieves a greater drop in blood pressure.

The ICEA of Drug A vs. B is \$10 per mmHg drop in BP. This means that for each 1 mmHg drop in blood pressure you achieve with Drug A (over and above what would have been achieved if drug B were taken) you will pay \$10.

PHARMACOECONOMIC METHODS: COST-UTILITY ANALYSIS

- Integrates patient preferences and quality of life when comparing treatment alternatives
- Used when quality of life is an important outcome of the drug in study
- Utility measured is often Quality-Adjusted Life Year gained (QALY)
- Results are expressed in ratio translated into cost per QALY gained → therefore the best alternative is the one with lowest cost per QALY gained
- Also common is the Incremental Cost Effectiveness Ratio (ICER) which measures the cost of 1 additional QALY (it answers the question, "how much more do I need to pay for each additional QALY gained through a superior therapy?")
- Cost Utility analysis is mostly used when comparing alternatives that extend life but have serious side effects (e.g. chemotherapy). Key advantage: benefits need not be expressed in dollars

PHARMACOECONOMIC METHODS – CUA

Cost-Utility Analysis - Practice question:

There are two new medications for the treatment of HIV.

- a) Using CUA, which drug is the best alternative?
- b) What is the Incremental Cost Effectiveness Ratio (ICER) of Drug B over Drug A?

Assume a 1-month study period

Drug	Administration cost	Drug cost/month	QALY gained
A	\$75	\$1400	0.25
B	\$0	\$1600	0.33

Answer:

a) Drug A: $(\$1400 + \$75) / 0.25 \text{ QALY} = \$5900/\text{QALY}$
Drug B: $(\$1600 + \$0) / 0.33 \text{ QALY} = \$4848 / \text{QALY}$

b) ICER of B over A = $\frac{\Delta \text{Cost}}{\Delta \text{QALY}} = \frac{\$1600 - \$1475}{0.33 - 0.25} = \$1,562.50/\text{QALY}$

Therapy with Drug B has the lowest cost per QALY gained and therefore is the best to use

Therefore, with Drug B you get additional QALYs costing you \$1,562.50 per QALY

PHARMACOECONOMIC METHODS: SUMMARY

Pharmacoeconomic methods	Cost units	Benefit Unit	Comparator Units	Which to choose?
Cost-Benefit Analysis (CBA)	\$	\$	Unitless (i.e. \$B/\$C)	Highest B/C ratio
Cost-Utility Analysis (CUA)	\$	Quality Adjusted Life Year (QALY)	\$/QALY	Lowest \$/QALY
Cost-Effectiveness Analysis (CEA)	\$	Measurable clinical outcome (e.g. additional life year, sugar level, LDL level)	Cost per measurable unit of benefit (units could be things like life years added, LDL levels lowered, sugar levels lowered)	Lowest Cost / Unit Benefit
Cost Minimization Analysis (CMA)	\$	N/A	\$	Lowest Cost

APPLICATIONS OF PHARMACOECONOMICS

- Clinical and policy decision making: one of the primary applications
 - Allowing practitioners and administrators to make better and more informed decisions regarding the products they offer
 - Considers economic, clinical, and humanistic outcomes
- Used to make formulary decisions
- Design disease management programs
- Measuring cost-effectiveness of interventions
- In industries, it is used in deciding among specific research and development alternatives
- In government, it is used in determining program benefits and prices paid
- In private sectors, it is used in designing insurance benefit coverage

INTERPRETATION: 11 CRITERIA FOR EVALUATING PHARMACOECONOMIC STUDIES

1. Objective: the purpose of the study should be given
 2. Perspective: researcher must list one or more perspectives
 3. Pharmacoeconomic method: CEA, CMA, CBA, or CUA
 - Their appropriateness for the given problem should also be assessed
 4. Study designs: prospective designs are preferred
 5. Choice of interventions: all relevant treatment options should be considered and included, their effectiveness should be established
 6. Costs and consequences: all the important cost/consequences of the treatment alternatives should be identified and should be relevant to the study perspective
 7. Discounting (discussed next slide)
 8. Results
 9. Sensitivity analysis (discussed next slide)
 10. Conclusions: must be justified and able to extrapolate to clinical practice
 11. Sponsorship – should be considered when evaluating study quality (e.g. any bias present)
- 

SENSITIVITY ANALYSIS (SA)

- Most analyses are based on assumptions. Usually, these assumptions consist of fixed numbers. But what if there is some uncertainty surrounding these assumptions?
- SA is a tool used to deal with uncertainty. E.g. a drug's effectiveness may vary by patient
- SA examines how study results change if the inputs are changed
 - Variables such as % effectiveness, incident of adverse effects, etc. are altered over a range of values (e.g. -15% and +15%)
 - Results are re-calculated to see if the conclusions change
- Allows practitioners to deal with data uncertainties and to appreciate how conclusions may differ by changing inputs

Example:

- Would agent A still be most cost effective if the effect of agent B was 20% greater than what was measured in the clinical trial?
- Would agent A be still most cost-effective if monitoring costs of Agent B was lowered?

SAMPLE QUESTION

A hospital pharmacist is conducting a Cost-Effectiveness Analysis (CEA) comparing two lipid-lowering agents. The initial base-case model concludes that Drug A is the most cost-effective option, costing \$40 per mmol/L reduction in LDL cholesterol. However, this baseline calculation assumes that patients will take the medication exactly as prescribed (100% adherence).

Recognizing that real-world adherence is rarely perfect, the pharmacist wants to know if the conclusions will change if patients miss doses. The pharmacist alters the input variables and recalculates the "\$ per mmol/L reduction" using estimated adherence rates of 80%, 60%, and 40% to determine if Drug A still remains the superior economic choice under these different conditions.

What specific pharmacoeconomic technique is the pharmacist applying to deal with this data uncertainty?

- a) Incremental Cost-Effectiveness Analysis (ICEA)
- b) Sensitivity Analysis (SA)
- c) Discounting
- d) Cost-Minimization Analysis (CMA)

DISCOUNTING (TIME VALUE OF MONEY)

Discounting – also known as the time value of money:

- For valid conclusions regarding costs generated over time, it is important to know that money delivered at different times has a different value
- Money in hand is worth more than the same amount in the future

What type of analysis should be used?

- a) A provincial health authority is comparing a vaccination program a new smoking cessation clinic considering its overall benefits and cost.
- b) A hospital committee is deciding between a new IV antibiotic and a standard oral antibiotic for pneumonia. The IV antibiotic cures patients faster but costs more. With respect to clinical outcome, the committee is focused on "days to clinical cure".

CHANGE LOG

August 2025

- Content reviewed, no changes made

October 2025

- Content reviewed. Improvements made to slide 17 with no change in meaning (so reprinting is not necessary)

March 2026

- Added Slide 31 (Sample Sensitivity Analysis Question)